Spatial-Temporal Crime Prediction and Socioeconomic Modeling in San Francisco (2020-2024)

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Website: https://swikritijoshimalla.github.io/SFPD_2020-2024/

Video: https://youtu.be/NT5tgeAKR_U

Assigned Tasks in This Project

Eddie Rosas was responsible for the Data and Methods section, including data processing, environment and tools setup, spatial data preparation, and dataset splitting. He led the Feature Engineering work, conducted Time-Series Forecasting using Prophet and SARIMAX, and developed the Dashboard and Visualizations. Eddie also contributed to Exploratory Data Analysis, collaborated on writing the Conclusion, and produced and edited the project Recording. Eddie reviewed and refined the final report to ensure clarity, coherence, and accuracy.

Swikriti Joshi was responsible for website creation; she designed and formatted the project site using HTML, CSS, and GitHub for hosting and version control. She also contributed to the project by drafting and editing the Introduction, completing the Literature Review, and writing the Discussion section, including insights, implications, limitations, and future work. In addition, she developed the SWOT Analysis and co-authored the Conclusion section.

ABSTRACT

This study analyzes spatial and temporal crime patterns in San Francisco using incident-level data from 2020 to 2024. Daily crime counts were utilized to train a Decision Tree classifier that differentiates between high-incident and low-incident days, with model performance assessed through TimeSeriesSplit cross-validation (score = 0.61). Monthly aggregated crime data facilitated short-term forecasting through SARIMAX and Prophet models, which estimate 30-day crime trends at the city level. Spatial preprocessing included aggregating incidents by police district and linking results to official district boundaries, enabling district-level mapping and visualization. These spatial and temporal analyses were integrated into an ArcGIS Online dashboard that displays multi-year crime trends, district-level activity, population indicators, and forecast summaries. Collectively, the classification model, forecasting results, and geospatial visualizations establish a comprehensive framework for analyzing crime dynamics in San Francisco and illustrate how structured analytical workflows can inform data-driven public safety planning.

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1. Introduction

This project aims to build and evaluate crime forecasting models for San Francisco and turn them into an interactive dashboard that can support more proactive, transparent, and equitable public safety planning. The main goal is to understand how crime has changed across the city from 2020 to 2024 and to make those patterns easier for both the public and decision-makers to explore. Instead of relying on assumptions or past memory, the project uses real data to answer practical questions such as: When are high-crime days most likely to occur? Which police districts tend to face more risk? And how do weekly routines, seasonal trends, and neighborhood differences shape crime levels?

To do this, we worked with five years of detailed police incident records and used three main approaches: machine learning, time-series forecasting, and geospatial mapping. In Python, we trained a Decision Tree model to classify days as “high-incident” or more typical, based on factors like recent crime activity, the day of the week, and the police district. The model performed reasonably well and, more importantly, provided clear and easy-to-understand rules about when crime risk tends to increase. For short-term forecasting, we built two models: Prophet and SARIMAX, to predict how many crimes might occur citywide over the next 30 days. Prophet did especially well at capturing weekly patterns, such as higher crime toward the end of the week, and produced forecasts with low error rates. While these models cannot predict exact events, they give a realistic picture of what the near future may look like if trends continue. We also examined crime from a spatial perspective by linking incident data to official police district boundaries. This makes it possible to compare districts, see which areas consistently experience higher activity, and understand how risk is distributed across the city rather than focusing only on total counts. All the results: maps, charts, model insights, and forecasts, were brought together into an ArcGIS dashboard. The dashboard is fully interactive and built so that both community members and public safety staff can explore crime patterns without needing technical skills.

Throughout the project, we focused heavily on fairness, transparency, and responsible communication. Because crime data can contain reporting biases and may not reflect all incidents equally across neighborhoods, the models are treated as decision-support tools rather than absolute predictions. The goal is to help agencies plan more thoughtfully, such as improving patrol scheduling, community outreach, or prevention programs, while avoiding harmful or unfair use of predictive analytics. By combining classification models, forecasting methods, and spatial dashboards in a clear and explainable way, this project provides a solid and repeatable framework for understanding crime dynamics in San Francisco and supporting more informed and equitable public safety strategies.

2. Literature Review

2.1 San Francisco Crime Classification

Link: https://scottmduda.medium.com/san-francisco-crime-classification-9d5a1c4d7cfd

San Francisco Crime Classification by Scott Duda describes a project that uses machine learning to predict crime categories happening in San Francisco based on when and where a crime happens. The idea behind the project is that police departments collect huge amounts of crime data over many years, and this information can be used to understand patterns and make better decisions. The authors focus on a large dataset from the San Francisco Open Data portal, originally used in a Kaggle competition, which contains almost 12 years of crime incidents from 2003 to 2015. The team started by exploring and cleaning the data. They removed duplicate rows and fixed errors such as coordinates that pointed to Antarctica instead of San Francisco. They removed features that were not available in the test set like Descript and Resolution that could not be used to build a fair model. After cleaning the data, they engineered many new features to help the model learn patterns. They created time-based features like year, month, hour, whether the incident happened on a weekend, and whether it occurred at night. They also built geographic features using the X and Y coordinates, including transformations, rotations, and even clustering to group similar locations. One important part of their feature engineering was using Word2Vec to convert crime addresses into numerical vectors. This allows the machine learning algorithms to understand location patterns in a more meaningful way than simply using text. They also created "log odds" features, which represent the likelihood of each crime type based on certain factors like hour of the day, day of the week, and address patterns. After preparing the data, the authors trained several types of machine learning models. These included tree-based models like LightGBM, Random Forest, and CatBoost, as well as a fully connected neural network. They evaluated each model using 5-fold cross-validation to make sure results were reliable. To improve accuracy, they used a stacked ensemble, which combines predictions from several different models. This final ensemble performed better than any single model and would have ranked in the top 40% in the original Kaggle competition. This article shows how a real-world crime prediction problem can be approached step-by-step using data cleaning, feature engineering, machine learning, and ensemble modeling. It also highlights the importance of using a wide range of features, time, location, clustering, embeddings to capture the complexity of crime patterns in a large city.

2.2 Crime Classification and Prediction in San Francisco

Link: https://cs229.stanford.edu/proj2015/254_report.pdf

The notebook Crime Classification and Prediction in San Francisco by John Cherian and Mitchell Dawson walks through how San Francisco crime data can be cleaned, explored, and modeled using different machine-learning and forecasting methods. The authors start by preparing the raw SFPD data, fixing missing fields, and organizing the information so it can be used in analysis. They then look at how crime changes over time, by day, by month, and by district to get a sense of the patterns that repeat from year to year. After the exploratory work, they build several models, including Decision Trees to classify high-crime days and forecasting tools like Prophet and SARIMAX to predict short-term crime trends. The notebook also explains how they evaluate the models by checking accuracy, reviewing the most important predictors, and comparing predicted results with what actually happened.

This approach closely relates to our capstone project, which uses many of the same tools and ideas. Like the notebook, we work with SFPD's incident-level data and use machine learning to identify when crime levels are likely to increase. We also analyze district-level trends and look at how crime changes by time of day, day of the week, and season. The difference is that our project focuses strongly on turning these analyses into something that can be used by the public and by city planners, not just data scientists.

While the notebook mainly shows how models are built and tested, our capstone extends the work by translating these results into interactive visual tools. By placing everything into ArcGIS map dashboard, we make the crime patterns easier for people to explore, whether they are community members, analysts, or city decision-makers. Our goal is similar to the authors', but broader: we want the predictions and visualizations to support more transparent, informed, and equitable public-safety planning. In simple terms, the notebook builds the models, and our capstone shows how they can be used in real-world decision-making.

3. Data and Methods

3.1 Data Sources

The Incident Reports from the San Francisco Police Department served as the base dataset and were downloaded from the San Francisco Open Data Portal. This dataset includes incident-level crime information with the date of an event, category of incident, location coordinates, and police district. The records for the period beginning from 1 Jan 2020 to 31 Dec 2024 were downloaded in order to make the observation temporal window uniform within the five-year period. The current period has been chosen to provide enough historical depth for time-series analysis while minimizing changes in reporting patterns.

The geometries of police district boundaries, obtained from an authoritative city shapefile, served as the spatial container for collating incident data. The boundary file functioned as the spatial layer for visualization and dashboard integration. However, incident district assignments were determined using the district attribute contained within the SFPD data. These polygons were used to compute district-level summaries, to generate geospatial feature layers, and to publish aggregated results to ArcGIS Online for mapping and dashboard visualization.

Although socioeconomic datasets such as the American Community Survey (ACS) were considered during the conceptual phase, external demographic variables were ultimately excluded from the final modeling pipeline. The scope of the analysis was intentionally restricted to incident data and district boundaries to maintain a clean methodological pipeline and to avoid assumptions introduced by merging multilevel population data sets. Therefore, any spatial–temporal results obtained necessarily characterize the crime dynamics as it unfolded within the police district organizational framework, and only there.

The incident dataset and district boundary file taken together formed the complete data environment for the study. The nature and quality of these sources enabled us to develop predictive models, carry out classification tasks at the district level, and create a geospatial dashboard-all under one common methodological umbrella.

3.2 Preprocessing

A staged process of data processing and feature extraction (cleaning, standardizing, and transforming) was employed to transform incident-level crime records into datasets that were applicable for spatial analysis, classification, and time-series prediction. The original dataset was taken from San Francisco Police Department Open Data Portal, and consisted of event-level incident records, which were later aggregated into daily counts for modeling. Entries that were missing, incomplete, or incorrectly formatted were discarded to maintain in integrity of the data for the following analytical stages.

Incident data for the temporal component was compiled into a daily time-series dataset covering 2020 to 2024. Such a format was fed as input to forecasting models, which expect regularly spaced observations rather than the discrete event-level logs. For classification, the district-level daily incident counts were aggregated and transformed into a binary target, identifying days with high incidents and the days with low incidents according to the thresholds derived from the distribution. In the forecasting process, external regressors were excluded, and the models utilized only the intrinsic temporal patterns present in the daily crime counts.

Spatial preprocessing involved aggregating incidents at the police-district level, followed by joining these results to district boundary polygons in ArcGIS Online for visualization. A reduced dataset covering the years 2022 to 2024 was published as a feature layer in ArcGIS Online to enhance dashboard performance and maintain critical spatial-temporal patterns.

Together, these preprocessing processes formed a harmonized and reproducible building block analytical pipeline that all forecasts, classifications and spatial outputs are embedded in, methodologically.

3.3 Spatial Data Preparation

Spatial data preparation involved transforming crime incident records into a format suitable for district-level analysis and visualization. Police district boundaries serve as the primary spatial unit, as they provide a stable organizational framework for incident records. Instead of using a coordinate-based spatial join, individual crime events were assigned to districts based on the Police District field present in the SFPD dataset. This approach ensured that temporal summaries were accurately linked to their respective district boundaries, facilitating the identification of persistent hot spots and district-level variations.

After aggregation at the district level, the datasets were published as hosted feature layers in ArcGIS Online, with district boundary polygons serving as spatial containers for map rendering. These spatial layers enabled interactive visualization, district-level comparison, and integration with dashboard indicators, allowing users to apply temporal filters and utilize analytic tools within the ArcGIS environment. As the study focused solely on crime distributions and excluded census-tract or socioeconomic covariates, the spatial routines prioritized geometric consistency and interpretability over multilevel demographic modeling. This approach resulted in a reproducible spatial pipeline appropriate for both exploratory spatial data analysis (ESDA) and final dashboard deployment.

3.4 Tools and Environment

For this project, we used Jupyter Notebook and Python as our analytical environment because they support data processing, machine learning, forecasting, and geospatial visualization. Jupyter Notebook was chosen for all preprocessing, modeling, and evaluation tasks because it is a stable platform for package management and reproducible computation. All data manipulation was done using pandas and numpy libraries to clean, aggregate, and transform incident-level crime records into a district-day dataset for classification and forecasting.

This environment was also used for machine learning workflows, which relied on libraries such as scikit-learn. This library included Pipeline and ColumnTransformer to structure preprocessing, encode categorical variables, and standardize numerical features. Hyperparameter optimization was performed using GridSearchCV, RandomizedSearchCV, and TimeSeriesSplit, ensuring that the model tuning respected the temporal ordering of the data and preventing information leakage.

Extending these analytical capabilities to forecasting, we utilized Prophet and SARIMAX. Prophet allowed us to model longer-term trends, weekly cycles, and yearly seasonality in daily crime counts, while SARIMAX served as a classical benchmark for capturing autoregressive and seasonal dependencies. Together, these tools enabled evaluation of short-term forecasting performance.

The geospatial visualization and dashboard were created in ArcGIS Online, which already included San Francisco district-level layers. This helped us create an interactive platform for mapping crime trends, predicted risk levels, and short-term forecasts. The integration of Python with the ArcGIS platform enabled an easy workflow for generating, analyzing, and communicating spatial and temporal insights.

3.5 Train/Validation/Test Split

The TimeSeriesSplit method was used during cross-validation to maintain temporal ordering and avoid information leakage during Decision Tree classifier tuning. Several forward expanding folds were used to learn and evaluate the model, simulating a scenario where time matters in testing model performance. Hyperparameter tuning was performed using cross-validation. For final model development, data were divided chronologically into 80% for training, 10% for validation, and 10% for testing. The SARIMAX and Prophet models were applied to the entire daily time series for forecasting purposes. Forecasts were assessed visually rather than with holdout-based scores, which is consistent with the exploratory nature of the forecasting module.

3.6 Database Management and Big Data Analysis Methods

The project required a systematic data processing and methodological approaches for preparing the San Francisco Police Department Incident Report dataset for analysis. Crime information was obtained from the City of San Francisco Open Data Portal, a well-known public data repository that has been used extensively in academic and policy research. The dataset includes detailed records of incidents with spatial and temporal information, making it suitable for exploratory analysis, prediction, and mapping applications.

To ensure consistency and reproducibility, a structured directory system was implemented to separate raw, interim, and processed data:

/bda600-capstone/data/raw

/bda600-capstone/data/interim

/bda600-capstone/data/processed

Python was used as the main working environment for data preparation. Pandas and

NumPy were used to clean, aggregate and transform time-series data. District assignment relied on the existing "Police District" field in the dataset, while ArcGIS Online was used to join aggregated district-level data to district boundary polygons for visualization. We used scikit-learn pipelines for the preprocessing and running of the Decision Tree classification. Prophet and SARIMAX were chosen to explore temporal forecasting because crime exhibits strong weekly cycles, yearly seasonality, and long-term temporal patterns. ArcGIS Online was used as the platform for visualizing district-level trends using the 2022 - 2024 data subset, supporting interactive mapping and integration with dashboard indicators. These tools and protocols created a structural and transparent workflow that allowed the data to be investigated, visualized, and interpreted in a reproducible manner suitable for validation and future extension.

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) facilitated the characterization of San Francisco crime by examining temporal (daily) and geospatial (district-level) incident trends from 2020 to 2024. Analysis of the five-year dataset revealed substantial variation, with reduced activity observed in early 2020 and gradual increases in subsequent years. However, these trends varied among districts. Consistent hotspots were identified, whereby some districts had higher incident counts for all years. The application of a five-year observation window reduced the impact of short-term fluctuations and yielded a more stable representation of long-term crime patterns. These results served as a baseline for the subsequent modeling steps by highlighting persistent district-level differences and recurring periods of elevated crime.

5. Feature Engineering

5.1 Label Definition

The classification task involved predicting a binary target variable indicating whether a given district-day was a high-incident day. The daily crime numbers were derived from the raw incident reports. A district-day was classified as high-incident (*y*_high = 1) when its daily count surpassed the district's rolling 75th-percentile threshold during the 2020 to 2024 period. This percentile-based cutoff provided a robust and easy-to-interpret method for detecting relatively high-activity days of data. Lag features (lag1, lag7), weekly indicators, and calendar variables were constructed to represent short-term persistence and weekly seasonality in crime patterns. The classification model employed a multivariate approach, integrating temporal lag features, calendar indicators, and categorical variables including police district and incident category. This design facilitated the development of an interpretable baseline classifier and minimized unnecessary model complexity.

5.2 Feature Engineering Details

We created a series of temporal, calendar-based, and categorical features designed to capture daily crime patterns in San Francisco. The temporal features, including lag1 and lag7, represent the incident count from the previous day and the same day in the previous week, respectively. These features incorporate short-term persistence and weekly periodicity into the model, reflecting the tendency of crime to cluster in recurring daily and weekly patterns. We incorporated calendar features to model the effects of routine activities. These features included indicators for month, day of week, weekend, and holidays. Such variables capture seasonal fluctuations. They also highlight differences between weekdays and weekends, and account for the impact of public holidays on crime volume throughout the year.

Categorical features, such as Police District, captured spatial and crime-type heterogeneity and provided geographic context. These features accounted for persistent differences in crime levels across neighborhoods. The Incident Category feature represented the type of reported offence, enabling the model to distinguish temporal patterns across different crime categories. This project also explored socioeconomic variables, but they were not included in the final model. This decision maintains methodological clarity and reduces the risk of introducing demographic bias. As a result, our final model uses only temporal, calendar, and categorical features directly from the incident data.

5.3 Modeling Pipeline

The classification component of the project was designed to predict whether a particular police district would experience high-incident activity on a specific day. All of the scikit-learn pipelines were completed and include a columnTransformer for correct preprocessing. This involved the numerical structure, imputation, and standard scaling. Instead, category features were more frequently imputed and one-hot encoded, which allowed the Decision Tree Classifier to process all variables independently and at the same time, in a single processing step. The TimeSeriesSplit method was employed during cross-validation to maintain chronological order and prevent temporal leakage during model tuning. Three hyperparameter search strategies were used to achieve a reliable configuration. GridSearchCV, RandomizedSearchCV, and Bayesian Optimization were employed as complementary hyperparameter search strategies. Each method identified similar optimal model configurations. The model optimized using GridSearchCV was chosen as the final classifier due to its consistent performance across both validation and test sets.

5.4 Hyperparameter Tuning

The combination of GridSearchCV and TimeSeriesSplit ensured that model evaluation maintained chronological order and minimized the risk of information leakage. This approach guarantees that the model is trained exclusively on historical data and evaluated using subsequent data, thereby reflecting realistic prediction scenarios and mitigating the risk of data leakage. The evaluation framework ensured that the tuned model generated realistic out-of-sample predictions.

This grid search included interpreted complexity control options, such as entropy-based splitting, a maximum tree depth of 14, and a minimum of 25 samples to split an internal node (see Table 1). The minimum number of samples per leaf was set at 40, and pruning was performed with a low cost-complexity penalty.

Hyperparameter	Value
Criterion	entropy
max_depth	14
min_samples_split	25
min_samples_leaf	40
ccp_alpha	0.0001

Table 1. Tuned Hyperparameters and Cross-Validation Scores

The optimized model attained an accuracy of 0.624, a Macro F1 score of 0.612, and a ROC-AUC of 0.655 on the validation set. The calibration curve demonstrated that the model's predicted probabilities closely corresponded to the observed event frequencies (see Figure 1).

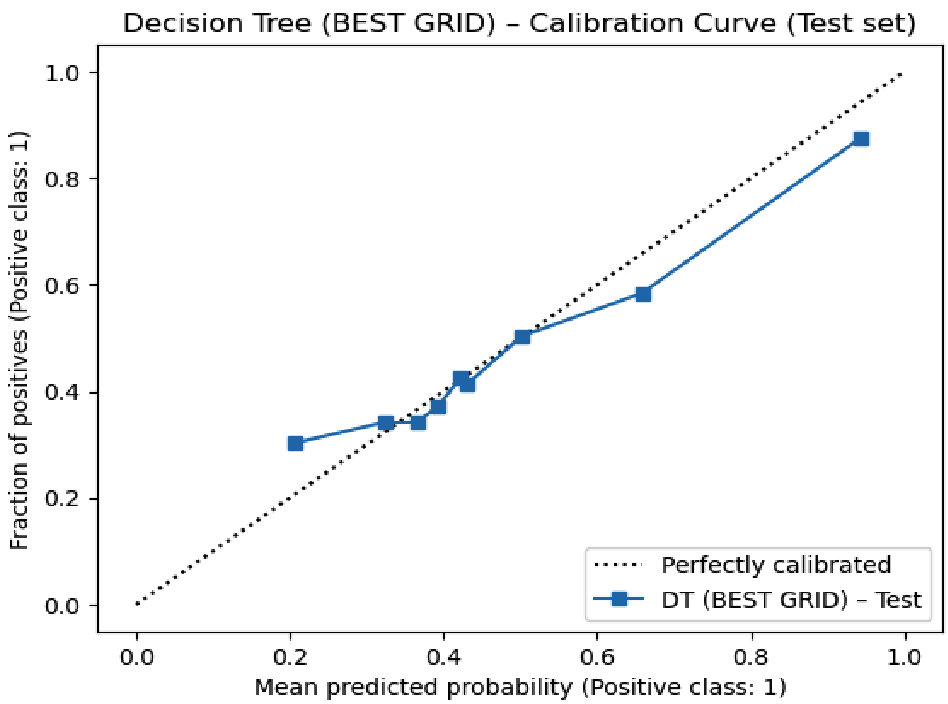

Figure 1. Calibration curve for the tuned Decision Tree classifier (BEST GRID).

These results suggest that the hyperparameter search avoided underfitting and overfitting by performing a balanced level of tree complexity.

5.5 Model Evaluation

The model performance in the hold-out test set was measured in a Macro F1 ≈ 0.61, Accuracy ≈ 0.63, and ROC-AUC ≈ 0.65 (see Figure 2). This means that the model performs pretty well in detecting minority classes in general and consistently across districts.

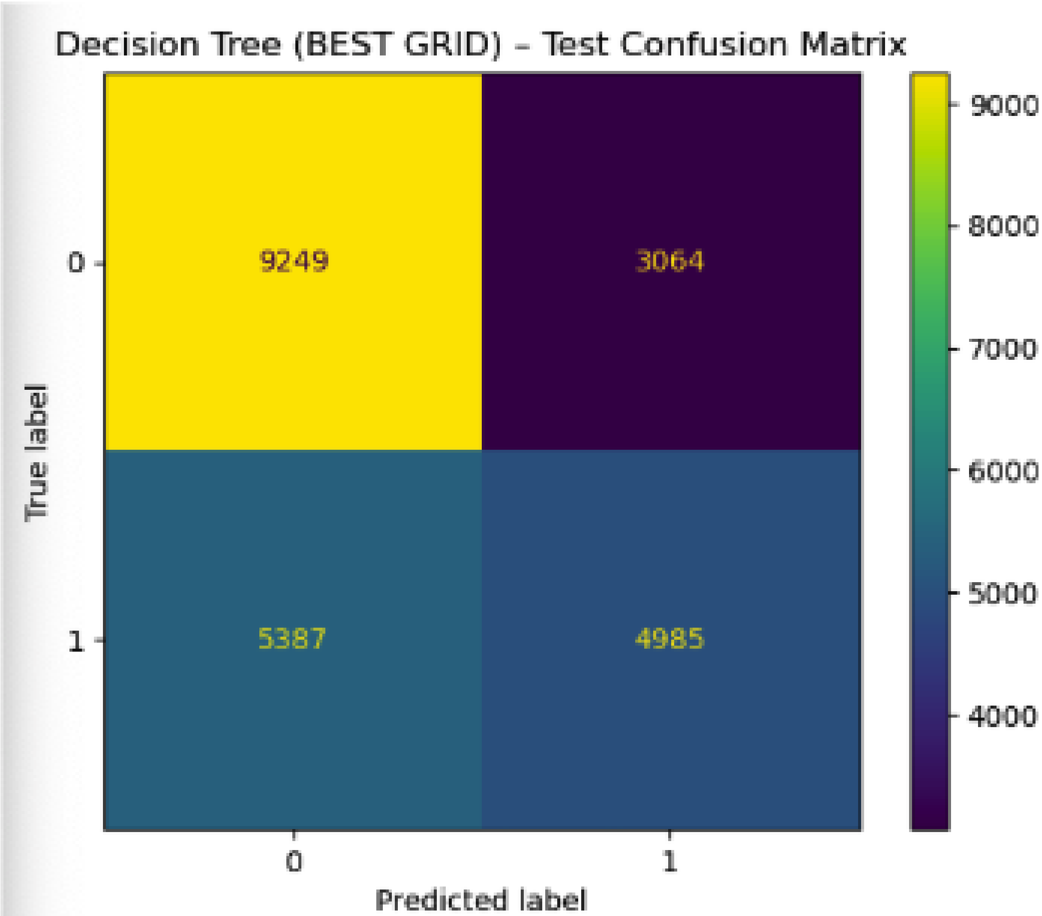

Figure 2. Confusion matrix for the tuned Decision Tree classifier (BEST GRID) on the test dataset.

The Precision-Recall curve demonstrates that probability estimates are calibrated (see Figure 3).

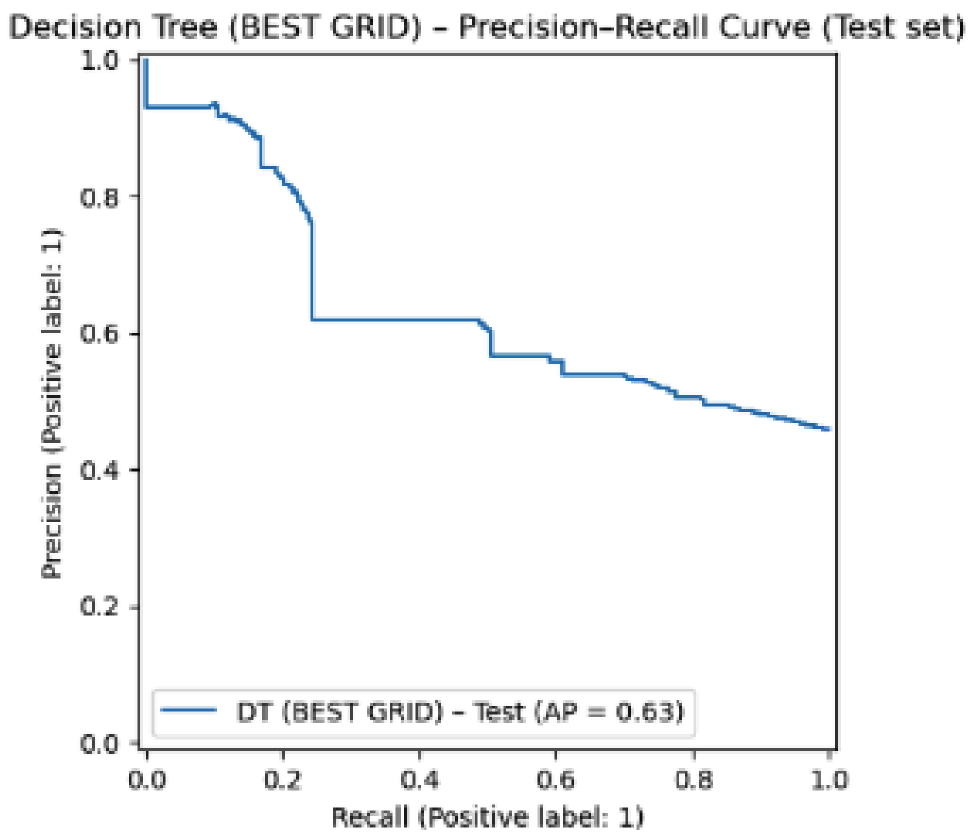

Figure 3. Precision–Recall curve for the tuned Decision Tree classifier (BEST GRID) on the test dataset.

A Confusion Matrix was used to test the classification results on unobserved data to understand better why the tuned classifier behaved differently in different cases. The confusion matrix presented the model's true positives, true negatives, false positives, and false negatives in the context of predicting high-incident days.

The ROC curve indicates moderate discriminative performance, consistent with the model's overall accuracy and F1 scores (see Figure 4).

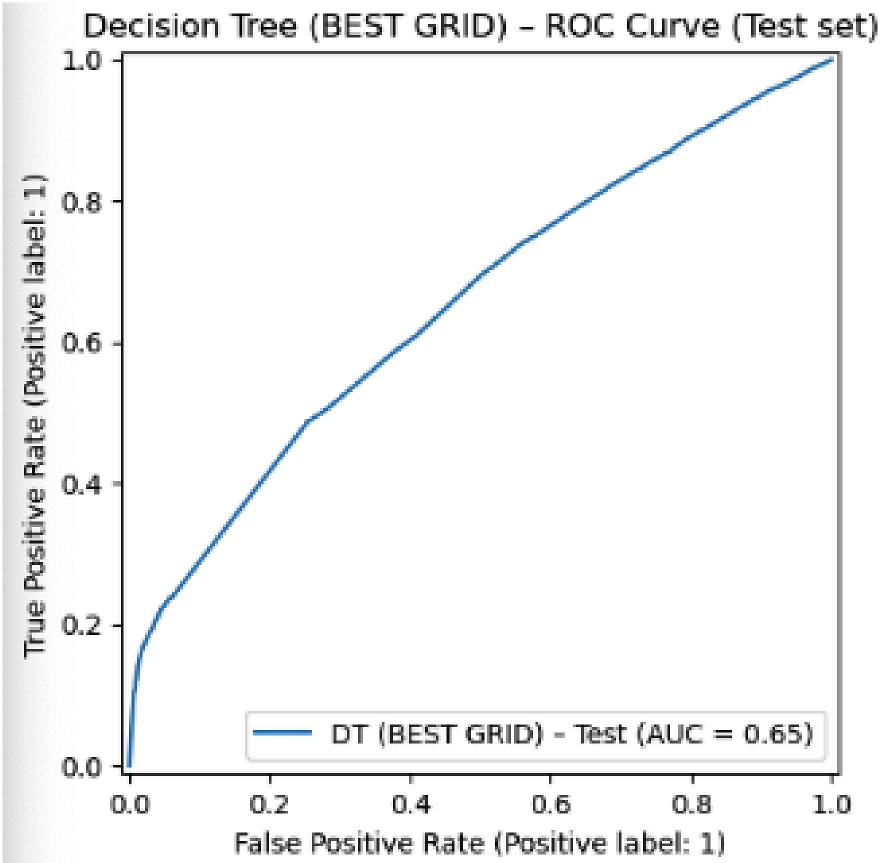

Figure 4. ROC curve for the tuned Decision Tree classifier (BEST GRID) applied to the test dataset (AUC = 0.653).

These visual performance summaries are more interpretable than they can be based only on numerical accuracy scores.

Model interpretability analyses indicated that lag1 and lag7 were the most influential predictors, highlighting the short-term persistence and weekly cycles characteristic of crime patterns (see Figure 5).

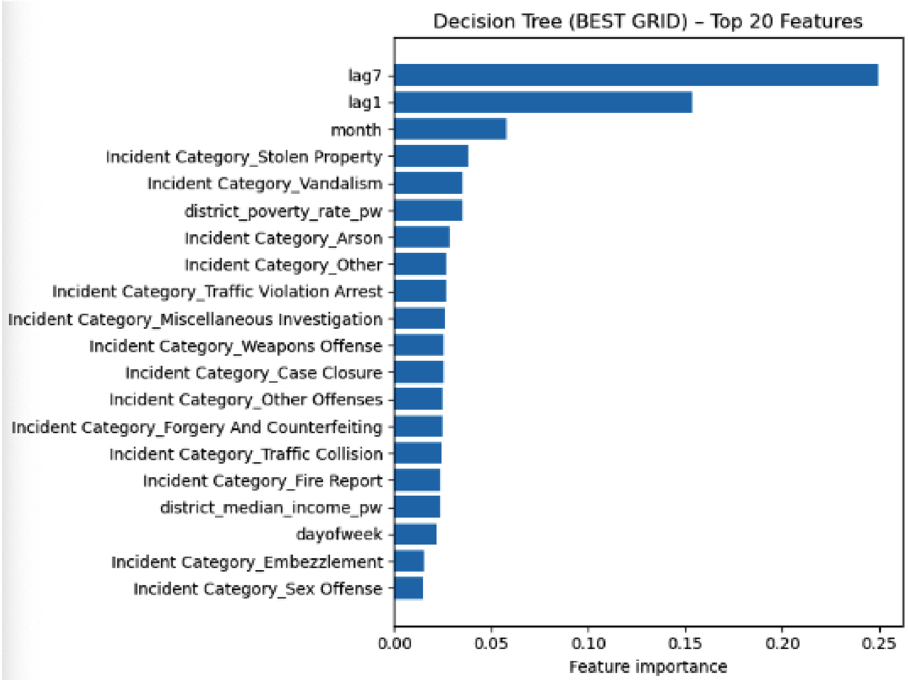

Figure 5. Top 20 feature importances for the tuned Decision Tree classifier (BEST GRID).

The model structure includes spatial context, such as police districts, and behavioral variables, such as weekends or holidays. Additionally, the attached file consists of a feature importance visualization, with these predictors among the top contributors.

They also have the potential for transparency and responsible analytics as a reason to use Decision Tree Classifiers. That means it is a set of decisions that can be easily understood and communicated to public safety partners. Decision trees are capable of modeling nonlinear relationships without the need for complex transformations and can effectively handle mixed data types. So, strong interpretability complements decisions concerning equity and ethics at high stakes. While ensemble models may provide greater predictive accuracy, their use would decrease transparency and increase monitoring complexity, which conflicts with the project's interpretability objectives. This pipeline can be modular, and future experiments can be conceived with such models that are accountable. The final classifier, therefore, provides an interpretable and helpful approach to assessing city crime risk, informs the ArcGIS Dashboard strategy, and enables informed urban safety planning.

6. Time-Series Forecasting

In addition to crime classification, two time-series forecasting models, Prophet and SARIMAX, were used to predict citywide total daily crime counts in San Francisco.

6.1 Prophet Forecasting

In fact, the decomposed additive model, with a mean absolute error of 21.56, a root mean squared error of 27.83, and a 8.6% MAPE, is very strong, which is well below the 10% threshold considered particularly suitable for forecasting tasks. Nearly all of the temporal structures accurately depicted by Prophet were long-term trends, high weekly seasonality, and annual cycles. Its component decomposition continued to increase on Fridays and Saturdays, with summer and early fall increases, and overall crime declined by late 2024. These interpretable components validated temporal features previously detected before classification modeling, as well as lag1, lag7, and day-of-week effects, which support the strong temporal dependence of crime patterns reported in San Francisco. The smooth forecast curve and widening uncertainty intervals showed that Prophet is stable and reliable for short-run planning (see Figure 6). The Prophet model was applied to daily incident counts in San Francisco from 2020 to 2024. This 30-day forecast is a predictable cyclical calendar with weekly seasonality and a gradual decline over time. The predicted values also reflect historical variability, while the uncertainty intervals widen and overlap across portions of the forecast horizon.

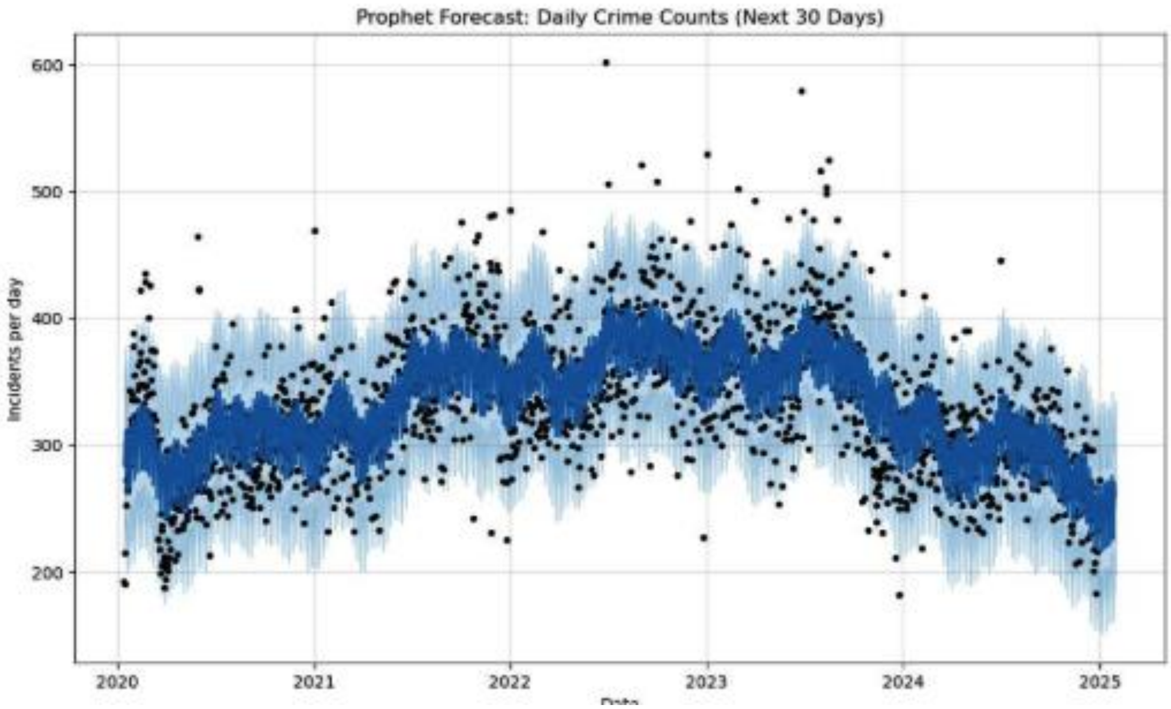

Figure 6. The plot shows the Prophet model's fitted values (blue line), predictive uncertainty intervals (shaded region), and observed daily crime counts (black dots). The model captures long-term trends, weekly seasonality, and year-over-year variation, and provides a 30-day forward projection.

6.2 SARIMAX Forecasting

The SARIMAX (1,1,1)(1,1,1,7) model served as an effective statistical baseline for short-term crime forecasting; however, its accuracy was lower than that of the Prophet model. For the 60-day forecast horizon, the SARIMAX model achieved an MAE 30.19, an RMSE of 37.09, and a MAPE of approximately 12.58%. The SARIMAX model forecast identified a weekly seasonal component with a period of s = 7, which aligns with established weekday and weekend crime patterns (see Figure 7). However, the confidence intervals were much larger than those of Prophet, suggesting that the short-run projections were much more difficult to predict. This was especially evident at localized peaks and troughs, where SARIMAX tended to smooth over sudden increases or decreases in crime volume. The model's structure produced a smooth forecast but tended to react slowly to abrupt changes in crime levels. The SARIMAX forecast captured the overall crime pattern but struggled with day-to-day variability caused by noise in the daily incident counts. These patterns align with the higher error metrics observed and illustrate SARIMAX's limited responsiveness to high-frequency fluctuations.

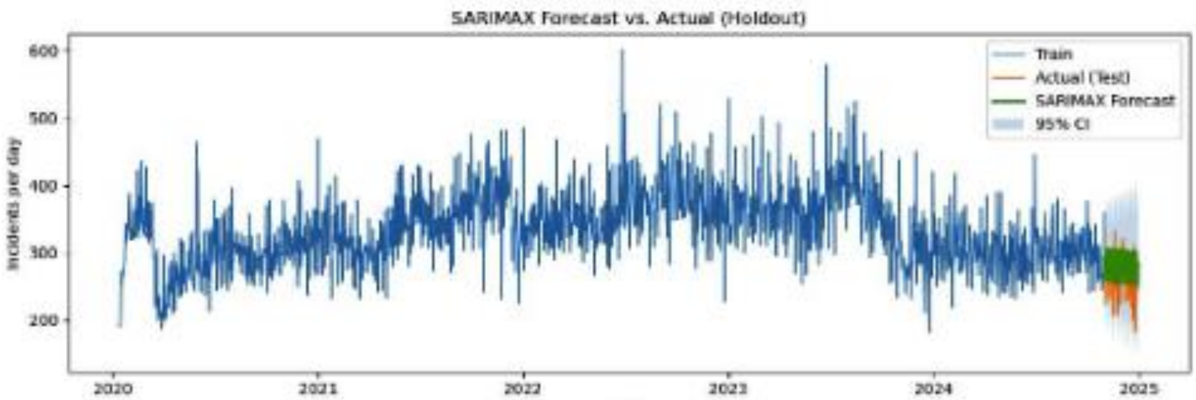

Figure 7. SARIMAX (1,1,1)(1,1,1,7) forecast for the 60-day holdout period compared with actual crime counts. SARIMAX captures the weekly seasonality but exhibits wider uncertainty bands and slightly higher forecast error than Prophet.

Residual diagnostics indicated that the SARIMAX model was appropriately specified and did not exhibit significant structural issues (see Figure 8). The residual histogram, which is essentially symmetric and has near-zero tails, indicates minimal systematic bias in forecasting. This does not mean that the model is always under- or over-predicting crime counts in the evaluation window. This argument is reinforced by the ACF plot of the residuals, which shows that most correlations between successive lags fall within the 95% confidence bounds (see Figure 9). This suggests that no significant remaining serial dependence remains unaddressed by the model. The SARIMAX model effectively captured the intended weekly seasonal pattern. The PACF plot confirmed that no significant partial autocorrelation remained in the residuals (see

Figure 10).

Collectively, the diagnostic charts indicate that the SARIMAX model effectively captured the primary temporal dependencies present in the data. However, strong statistical diagnostics alone did not result in optimal forecast performance. While the model successfully removed autocorrelation and seasonal dependence, its linear structure limit did not allow it to survive irregular, high-volatility movements in crime. Although the model demonstrated robust statistical properties, its dependence on linear dynamics resulted in slow adaptation to abrupt short-term fluctuations in crime levels, thereby reducing its effectiveness in forecasting highly volatile periods.

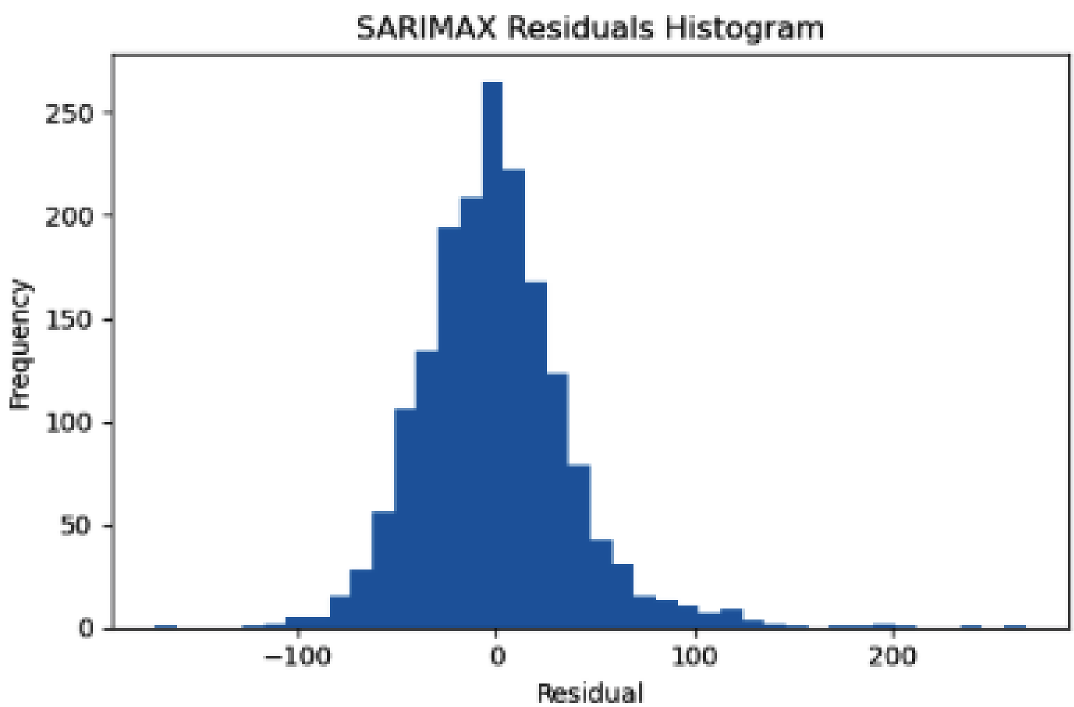

Figure 8. Histogram of SARIMAX residuals showing centered distribution and moderate tails.

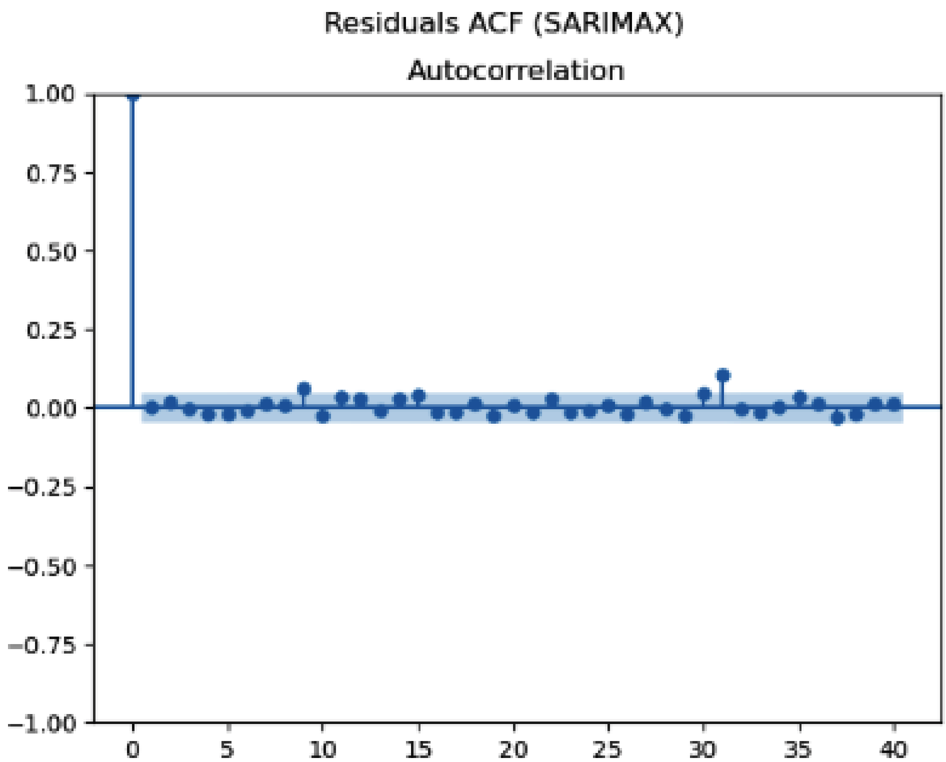

Figure 9. Autocorrelation function (ACF) of residuals. Most lag correlations fall within confidence bounds, indicating no major autocorrelation in the residual structure.

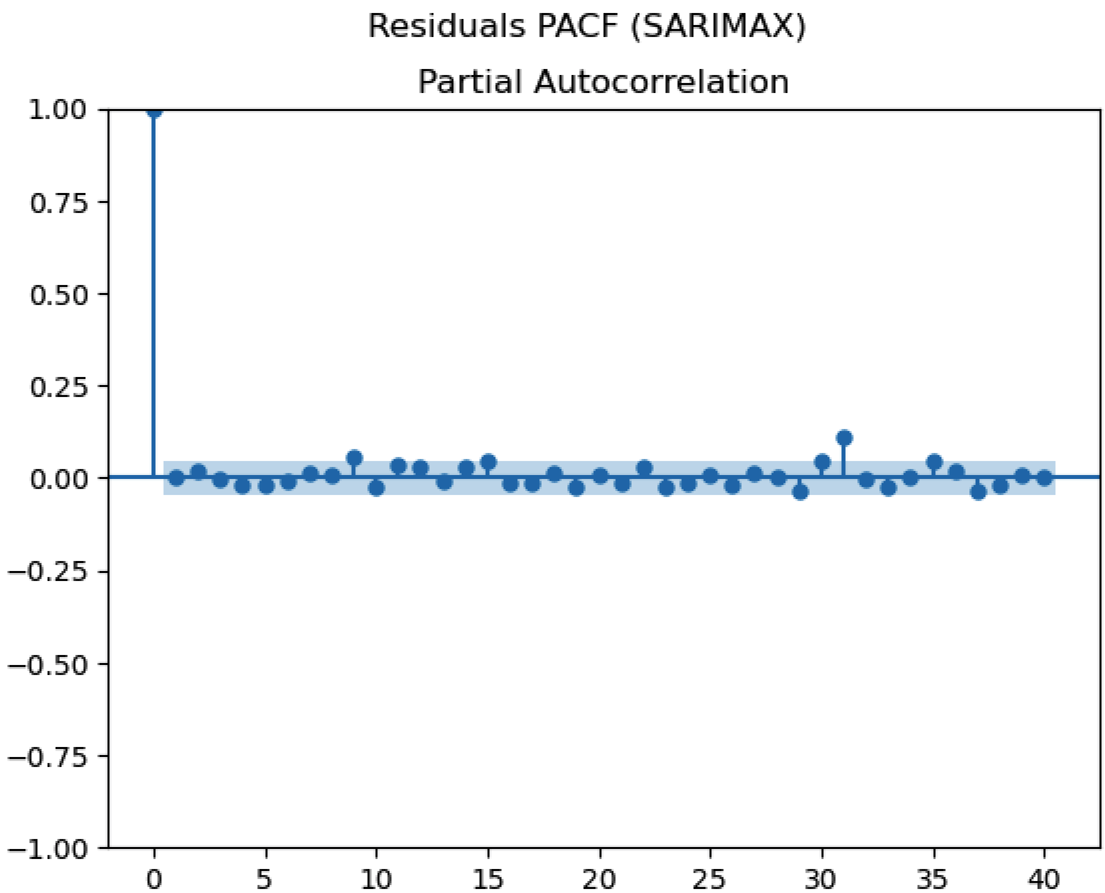

Figure 10. Partial autocorrelation (PACF) plot of SARIMAX residuals. The lag-1 spike is the only significant value, which suggests an absence of remaining partial autocorrelation in the residual structure.

6.3 Model Comparison and Evaluation

The performance differences between these two models are larger than previously apparent in the combined visual forecast analysis. Both models incorporated a weekly seasonal cycle; however, Prophet demonstrated this pattern with greater clarity in its forecast output. Prophet generated consistent short-term forecasts, providing stable daily estimates throughout the 30-day holdout period. Prophet's forward forecast curve was smooth, and its uncertainty bands widened gradually over the forecast horizon, reflecting increasing uncertainty farther into the future. In contrast, the SARIMAX forecast exhibited substantially wider uncertainty intervals and increased variability at turning points, which suggests reduced confidence in short-term predictions. This contrasts with the quantitative findings reported in Table 2, which showed that Prophet had significantly lower error metrics, including MAE (21.56 vs. 30.19), RMSE (27.83 vs. 37.09), and MAPE (8.58% vs. 12.58%).

Model	MAE	RMSE	MAPE
Prophet	21.56	27.83	8.58%
SARIMAX	30.19	37.09	12.58%

Table 2. Forecast Accuracy (Prophet vs SARIMAX)

The elevated SARIMAX error values further highlight the superior forecasting performance of Prophet. The differences show that Prophet did not just produce better point estimations but also provided tighter probabilistic bounds that are vital to operations forecasting. Based on the combined weight of statistics diagnostics, error measures, and quantitative visual comparison of the model used for the ArcGIS Dashboard, Prophet was chosen as the primary forecasting model for deployment on the ArcGIS Dashboard. Within the dashboard, Prophet's forecast was integrated into the summary indicators. In contrast, SARIMAX served mainly as a methodological benchmark rather than being directly visualized. The two models offered complementary insights during the analysis; however, Prophet was selected as the primary operational forecasting tool for dashboard deployment.

7. Dashboard and Visualization

[SF Crime Trends Dashboard (2022-2024)]

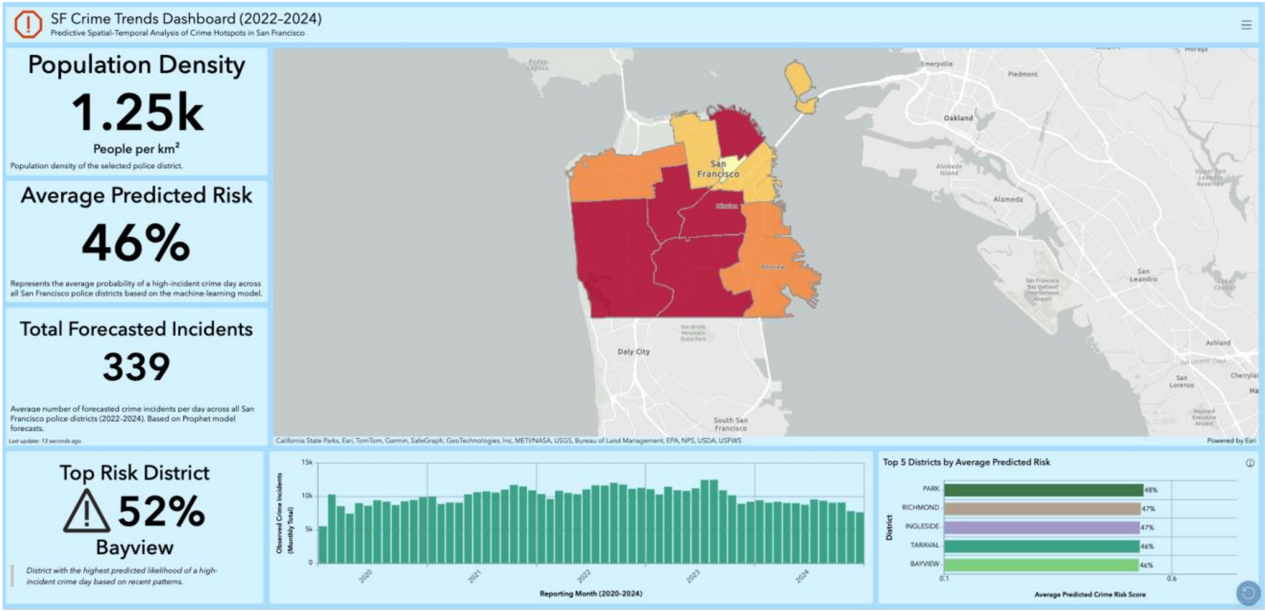

Figure 11: SF Crime Trends Dashboard

The SF Crime Trends Dashboard (2022-2024) is an interactive ArcGIS-based visualization designed to help users understand spatial and temporal crime patterns across San Francisco. Built to support predictive analysis and public awareness, the dashboard brings together several key metrics, geospatial layers, and trend indicators in a clean, organized layout.

At the top-left, the dashboard highlights Population Density, showing an estimated 1.25k people per km² for the selected police district. This gives immediate demographic context, helping users compare crime patterns between densely and lightly populated areas. Directly below this, the Average Predicted Risk panel displays a 46% probability of a high-incident crime day across San Francisco districts, summarizing the machine-learning model's forecast in an easy-to-read percentage.

The dashboard also features Total Forecasted Incidents, estimating 339 daily incidents across the city based on historical data and predictive modeling. These numerical indicators allow users to quickly grasp the overall crime environment before exploring deeper details.

A key highlight is the central map, which visually displays predicted crime risk using a color gradient. Districts shaded in darker red represent higher-risk areas, while lighter orange regions indicate lower relative risk. This map allows users to instantly identify spatial hotspots and compare risk levels across neighborhoods such as Bayview, Ingleside, and Park.

To the left of the map, the Top Risk District card identifies Bayview as having the highest predicted crime likelihood at 52%. Including both a percentage and a district name makes the information accessible and actionable, especially for public safety planning or community awareness.

Below the map, a monthly crime trend bar chart shows observed crime incidents from 2020 to 2024. This graphic reveals seasonal patterns, peaks, and declines over time. It helps users understand whether crime has been rising, falling, or remaining stable, and whether certain months consistently show higher activity.

On the right side, the dashboard presents a Top 5 Districts by Average Predicted Risk bar chart. Districts like Park, Ingleside, Taraval, and Bayview are compared side-by-side, with risk percentages ranging from 46% to 48%. This ranking helps users see not only which districts face the most risk but also how similar or different these values are from one another.

Visually, the dashboard uses light blue panels, clear typography, and structured grouping to make large amounts of information easy to read. The combination of numeric indicators, a geospatial map, and trend charts allows viewers to move seamlessly from high-level summaries to detailed insights. Overall, the dashboard provides an intuitive, data-driven overview of crime behavior in San Francisco, supporting both analysis and informed decision-making for public safety efforts.

8. Discussion

This project brings together classification models, time-series forecasting, and geospatial mapping to better understand how crime patterns in San Francisco have shifted from 2020 to 2024. By working with incident-level police data, grouping it into daily and monthly counts, and connecting it to police district boundaries, we built a framework that highlights when and where crime is more likely to occur. This section summarizes the main findings from the models, explains what they mean for planning and public safety, and reflects on important limitations and future improvements.

8.1 Insights from Machine Learning and Forecasting

The machine learning results show that crime in San Francisco follows clear time-based patterns. The Decision Tree model, trained using district-day crime counts, found that the strongest predictors were incident levels from the previous day and previous week (lag1 and lag7), day of the week, and police district. This means crime tends to follow weekly cycles and often continues patterns from the previous day. The model's performance, Macro F1 of about 0.61 and ROC-AUC of about 0.65, shows moderate but useful accuracy in identifying high-incident days. One of the benefits of using a Decision Tree is that it is easy to explain. The rules it generates can be interpreted in everyday language which makes the model more transparent for city agencies and easier to evaluate from an ethics standpoint.

The forecasting section focused on overall citywide crime counts. Both Prophet and

SARIMAX captured the strong weekly rhythm of crime, higher near the end of the week and lower midweek. Prophet outperformed SARIMAX with lower error rates, and it also provided clear visual components showing trends, weekly cycles, and seasonal changes. SARIMAX worked as a solid statistical baseline but struggled with sudden short-term changes. These results suggest Prophet is the stronger option for short-term operational forecasting, while SARIMAX remains useful as a reference model.

8.2 Implications for Urban Planning and Resource Allocation

The combination of classification, forecasting, and geospatial visualization produces a practical system that can support public safety decisions. The classifier identifies district-days with higher risk, which can help guide decisions such as when to increase patrols, schedule outreach teams, adjust staffing, or coordinate with community programs. If certain districts consistently experience spikes on weekend evenings, officials can prioritize lighting improvements, targeted patrols, or preventive community efforts.

On a citywide level, Prophet's 30-day crime forecasts provide a short-term view of expected incident levels. These projections can support planning for staffing, overtime budgeting, and resource allocation across units. Forecast uncertainty bands also help planners prepare for "typical days" and "higher-than-usual days."

The ArcGIS dashboard links these results to geographic boundaries, allowing users, from police analysts to community members, to explore district-level differences. Comparing predicted risk, total incidents, and time trends across districts helps highlight where crime is concentrated and where conditions differ from the citywide average. This supports more equitable planning by providing context, not just raw counts.

8.3 Limitations: Data Bias, Under-Reporting, and Timing Issues

Despite the strengths of the project, several limitations must be acknowledged. First, police incident data always reflects only reported crime, not all crime. Some categories, such as domestic violence or minor theft, are under-reported, and reporting rates vary by neighborhood and demographic group. This means the models capture reporting patterns, which may not perfectly match real-world conditions.

Second, some records lack exact coordinates or include only generalized locations, reducing the accuracy of hotspot mapping. Another challenge is that many incidents are assigned default times like 12:00 AM or 12:00 PM when the actual time is unknown, making hourly analysis unreliable. This is why the project uses daily aggregation instead of more detailed time breakdowns.

Third, the study focuses on police districts and daily granularity, which works for operational planning but may miss micro-level insights visible only at block or beat level. Adding socioeconomic and contextual variables could enrich the models, but also introduces the risk of embedding demographic bias if not handled carefully.

8.4 Future Work: Improved Models, Spatial Clustering, and Real-Time Updates

There are several opportunities to expand this project. Future work could include ensemble models that combine Decision Trees with gradient boosting or logistic regression to improve accuracy while maintaining interpretability. For time-series data, combining Prophet with deep learning models could help capture unusual spikes or sudden changes in crime levels.

Spatial clustering techniques, such as DBSCAN, HDBSCAN, or spatial scan statistics could identify smaller hotspot areas within districts and show how they change over time. This would help target interventions more precisely.

Another major improvement would be connecting the pipeline to real-time SFPD open data. This would allow the dashboard to update automatically and act as a live monitoring system. Real-time updates would also make it possible to continuously evaluate and improve the models as new data comes in.

This project shows that combining interpretable machine learning, forecasting models, and GIS visualization can create a useful and accessible system for understanding crime trends in San Francisco. Although there are limitations related to reporting, data quality, and granularity, the results provide a strong foundation for responsible, data-driven planning. With future enhancements, such as real-time updates, more advanced models, and finer spatial detail this framework can continue to support fair, transparent, and informed public safety decisions.

9. SWOT Analysis

Introduction to SWOT Analysis:

A SWOT analysis is a useful framework for evaluating a Big Data project because it breaks the work into four key areas: Strengths, Weaknesses, Opportunities, and Threats. Looking at these areas side-by-side allows us to understand not only what the project currently does well, but also where it may face limitations or require further refinement. The Strengths highlight the project's most valuable features and capabilities, while the Weaknesses identify challenges or gaps that may limit performance or reliability. Opportunities point to ways the project can expand, improve, or deliver even greater impact in the future. Finally, the Threats category draws attention to risks, such as ethical concerns, data issues, or public misinterpretation, that must be recognized and managed responsibly.

By examining the project from all four angles, we gain a more balanced and comprehensive understanding of its overall effectiveness. This holistic view helps guide decision-making, supports responsible use of data and technology, and ensures that improvements are aligned with the project's long-term goals. Ultimately, a SWOT analysis provides a clear roadmap for strengthening the project, maximizing its value, and addressing potential risks before they become problems.

Project SWOT Overview (Strengths, Weaknesses, Opportunities, Threats)

✅ Strengths	⚠ Weaknesses
• **Extensive Historical Dataset:** Uses five years of SFPD crime data (2020–2024), enabling identification of long-term trends, seasonal patterns, and neighborhood-level shifts. • **Comprehensive Workflow:** Combines Python for data cleaning and analysis, ArcGIS for mapping crime locations, and Tableau for interactive dashboards, creating a full end-to-end analytics pipeline. • **Actionable Visual Insights:** Dashboards highlight crime patterns by day of week and time of day, helping police plan patrols more effectively and allowing the public to stay aware of higher-risk times and locations. • **Accessible, User-Friendly Outputs:** Visual tools make complex analytical results understandable for both technical stakeholders and community members, improving transparency and engagement.	• **Limited Spatial Precision:** Many crime reports lack exact coordinates or only list nearby intersections or general areas, reducing hotspot mapping accuracy. • **Data Gaps & Inconsistencies:** Missing values and changes in reporting practices over time can affect the reliability of trend analysis and model performance. • **Time Recording Issues:** Numerous incidents are recorded at default times (e.g., 12:00 AM or 12:00 PM) when the exact time is unknown, which reduces the accuracy of time-of-day analysis for certain offenses.
📈 Opportunities	**⚠ Threats**
• **Policy & Strategy Integration:** Insights can guide staffing, patrol planning, and targeted prevention programs, supporting more efficient and equitable public safety strategies. • **Real-Time Expansion:** Connecting the framework to live SFPD feeds could enable near real-time crime tracking, alert systems, and faster response to emerging patterns. • **Community Awareness & Engagement:** Transparent dashboards and maps can help residents better understand safety trends in their neighborhoods and strengthen trust with local agencies. • **Advanced Predictive Modeling:** With improved data quality, the project could evolve into forecasting tools that anticipate high-risk times and locations for proactive intervention.	• **Ethical Risks in Predictive Policing:** If not carefully governed, predictive tools can unintentionally reinforce historical biases and disproportionately impact certain communities. • **Privacy & Data Security:** Sensitive geospatial and incident-level data must be protected with strong safeguards to prevent misuse and protect individual identities. • **Public Misinterpretation:** Hotspot maps and predictions may cause confusion or concern if limitations are not clearly communicated and contextualized. • **Dependence on Imperfect Data:** Incomplete or inconsistent data can lead to misleading insights unless these limitations are clearly acknowledged in all interpretations and recommendations.

Figure 12: SWOT Diagram

Strengths:

A main strength of this Big Data crime project is that it uses five years of detailed crime data from the San Francisco Police Department (SFPD). Having this much historical information makes it easier to see long-term crime trends, seasonal changes, and patterns that happen at different times of the year. Another strength is the full workflow used in the project, Python for cleaning and analyzing data, ArcGIS for mapping crime locations, and creating interactive dashboards. These tools make the results clear, accurate, and easy for both experts and everyday people to understand. Because the dashboards show crime patterns by day of the week and time of day, police can use this information to plan patrols better, and the public can use it to stay aware of riskier times and locations.

Weaknesses:

Even with its strengths, the project has some weaknesses. A major issue is that many crime reports do not include the exact location where the incident happened. Some reports only list a nearby intersection or a general area, which makes hotspot mapping less accurate. The dataset also has inconsistencies, such as missing information or changes in how crimes were recorded over the years. These gaps can affect how reliable the trend analysis is. Another limitation comes from how crime times are recorded, many incidents are entered as 12:00 AM or 12:00 PM because the exact time is unknown. When police pick whichever time seems closest, it reduces the accuracy of time-of-day analysis for certain crimes.

Opportunities:

This project offers many opportunities for improving public safety and community awareness. City leaders and police departments can use the insights to make smarter choices about staffing, patrol planning, and targeted crime prevention programs. Since the analysis highlights crime patterns by day and time, the public can also use this information to avoid high-risk areas during certain periods. In the future, the project could be expanded to connect with live SFPD data, allowing for real-time crime tracking and faster responses. The interactive dashboards and maps can also help build trust between communities and law enforcement by making crime information easier to understand and more accessible to everyone.

Threats:

There are also some risks that need to be carefully managed. Predictive policing tools may unintentionally repeat past biases if they are not used properly, which could unfairly target certain neighborhoods. Since the dataset includes sensitive location information, strong privacy protections are necessary to ensure the data is used responsibly. Another threat is the possibility that the public may misinterpret hotspot maps or predictions, leading to unnecessary worry or confusion. If the results are not clearly explained, people may misunderstand what the data is showing. , if the project relies too much on incomplete or imperfect data, the predictions or insights may not be fully accurate unless the limitations are openly addressed.

10. Conclusion

This study highlights the importance of developing an equitable machine learning framework that integrates spatial and temporal data, while conceptually considering socioeconomic context, to enhance the understanding and prediction of crime trends in San Francisco. The temporal dependencies observed in the crime data exhibited strong persistence and seasonality, as well as notable spatial variation among police districts.

While straightforward in performance and highly interpretable, the Decision Tree classifier performed well and was stable, with a macro F1 score of about 0.61 and an ROC-AUC score of about 0.65. These findings suggest that short-term criminal risk prediction is feasible with moderate reliability when employing a transparent rule-based approach. Additionally, feature importance analysis indicated that district-level contextual factors and routine-activity patterns, such as weekly and weekend cycles, were influential predictors in the model. The forecasting activity supplemented the classification task by demonstrating that Prophet outperformed SARIMAX for short-run forecasting, achieving a MAPE of approximately 8.6%. This resulted in the dashboard incorporating Prophet's forecast outputs into a spatially enabled decision-support system that enhances interpretation, transparency, and situational awareness over deterministic enforcement guidance.

This project also acknowledges the ethical risks inherent in predictive policing services, including reporting bias, misapplication of jurisdiction, and potential misuse of forecasts. For accurate communication of limits, the framework emphasizes its responsible and equitable use of findings. Although forecasting models are unable to predict abrupt structural shocks, the system illustrates the integration of data science, GIS, and ethical modeling principles into public safety planning.

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